Submission STAT1-MID-SUB02

Question STAT1-MID-Q01
work / setup:
1. x■=25/5=5.00
crossed-out arithmetic: [smudged]
FINAL (maybe): mean=5.00; sample variance=14.00 plus 1

Question STAT1-MID-Q02
work / setup:
1. P(D■∩D■)=(5/16)((5-1)/(16-1))
crossed-out arithmetic: [smudged]
FINAL (maybe): 0.0833 plus 1

Question STAT1-MID-Q03
work / setup:
1. SE=√(p■(1–p■)/n)=√(0.55(1–0.55)/40)=0.0787
2. CI=0.55±1.96(0.0787)=(0.396, 0.704)
crossed-out arithmetic: [smudged]
FINAL (maybe): (0.396, 0.704); repeated-sampling interpretation plus 1

Question STAT1-MID-Q04
work / setup:
1. Increase sample size n to reduce standard error.
2. Reduce confidence level or population variability when justified.
crossed-out arithmetic: [smudged]
FINAL (maybe): increase n; lower confidence level or variability; note the coverage trade-off plus 1

Question STAT1-MID-Q05
work / setup:
1. $0.03 < 0.05$, so reject H■ under the stated rule.
crossed-out arithmetic: [smudged]
FINAL (maybe): reject H0; evidence against H0, not proof that H0 is false plus 1

Question STAT1-MID-Q06
work / setup:
1. x■=35/5=7.00
crossed-out arithmetic: [smudged]
FINAL (maybe): mean=7.00; sample variance=4.50 plus 1

Question STAT1-MID-Q07
work / setup:
1. P(D■∩D■)=(5/19)((5-1)/(19-1))
crossed-out arithmetic: [smudged]

FINAL (maybe): 0.0585 plus 1

Question STAT1-MID-Q08

work / setup:

1. SE=√(p■(1–p■)/n)=√(0.35(1–0.35)/40)=0.0754
2. CI=0.35±1.96(0.0754)=(0.202, 0.498)

crossed-out arithmetic: [smudged]

FINAL (maybe): (0.202, 0.498); repeated-sampling interpretation plus 1